



Education

National University of Singapore (QS Top 10)

Aug 2025 – Jun 2027 (expected)

Master of Science in Biomedical Informatics

Zhuhai College of Jilin University (Rank: 1/235)

Oct 2020 – Jun 2024

Bachelor's Degree in Data Science and Big Data Technology

Professional Experience

Kriston.AI

Mar 2026 – Sep 2026

LLM Post-Training Engineer (Intern)

AI Algorithms Team 1 | Paediatric Sales Assistant

Post-trained a specialised LLM for multi-turn healthcare sales conversations focused on lead capture.

- **Training data and SFT.** Combined simulated user conversations with production logs; applied a shared quality rubric and balanced topics, dialogue lengths and out-of-distribution examples. Fine-tuned **Qwen3.5-27B with LoRA SFT** and brought the model to production following benchmark iterations and simulation-based acceptance testing.
- **Production feedback and distillation.** Used production traces to diagnose failures, repair responses and iteratively refine the model through SFT and DPO. Reduced the **evaluation failure rate from 25% to 5%** on a rolling set of 1,000 real conversations drawn from production traces. Trained topic-specific teachers and demonstrated the feasibility of **Multi-Teacher On-Policy Distillation (MOPD) into Qwen3.5-9B**. Outperformed previous internal paediatric LLM baselines across a three-tier evaluation framework covering launch readiness, production performance and iteration priorities.

AI Algorithms Team 2 | Unified Model across Medical Specialties

Contributed to a shared model for healthcare lead capture, progressively replacing specialty-specific models and proprietary model APIs.

Cold-Start Model and Continuous Improvement

- Contributed to **two-stage full-parameter SFT of Qwen3-8B-Base / Qwen3-8B-CPT**. Stage 1 used approximately **460,000 production logs** to construct complete multi-turn training conversations, teaching empathetic, goal-directed dialogue and standard consultation and lead-capture workflows. Stage 2 targeted final-turn responses in difficult examples and production failures, separating foundational dialogue training from targeted fixes.
- Built a **production-data feedback loop**: log ingestion, failure identification and root-cause analysis, response repair, quality checks, Stage 2 retraining and regression evaluation. Continuously adapted the cold-start model to real business contexts, user distributions and interaction patterns.

Agentic RL | Multi-Turn Reinforcement Learning

- **Environment and user simulation.** Recreated the production agent workflow so rollouts reflected real context, information access, interaction logic and business constraints. Built user personas and a simulator from real conversations and behavioural distributions, modelling evolving latent states, user feedback and stopping behaviour to improve environment fidelity.
- **Hierarchical rewards.** Designed trajectory-level and turn-level rewards combining overall conversation quality, final lead capture and changes in user state. Aggregated future process rewards using return-to-go and combined them with trajectory objectives to form per-turn policy advantages, addressing sparse rewards and credit assignment in long conversations.
- **Training and validation.** Customised **verl** for multi-turn rollouts and reward/advantage integration in a **GRPO** training loop. Audited post-training trajectories to identify and correct environment, simulator and reward biases across repeated training cycles. Raised internal **Excellent-Bench scores from 20% to 70%**, measuring alignment with expert sales behaviour, while keeping other evaluated capabilities broadly unchanged. The model entered a phased production rollout serving **100,000+ calls per day**.

 Professional Experience (continued)

Jiusong Health Technology

Jul 2024 – Oct 2025

AI Agent Application Engineer (Full-time)

- **Chronic-care agent development.** Co-developed and deployed a **WeCom** agent for chronic-disease follow-up, translating business requirements into an executable workflow with query rewriting, parallel multi-intent detection, hybrid retrieval and multi-stage generation. Supported diet, exercise, weight and blood-glucose management.
- **Domain-specific memory.** Curated real chronic-care conversations and applied **LoRA SFT** to resolve omitted context and implicit references that general-purpose LLMs struggled with in multi-turn dialogue. Deployed a domain-specific memory model with **vLLM** and integrated it into the agent's memory module, improving **key-information extraction accuracy from 85% to 95%**.
- **Proactive follow-up and persistent task memory.** Designed a follow-up calendar around standard operating procedures (SOPs), decomposing long-term care plans into tasks with trigger times, executable skills and tracked status. The harness used calendar state to trigger heartbeat events; the agent retrieved the current task, invoked the appropriate skill for data collection, metric analysis or proactive reminders, and wrote back results and patient state to keep the plan current.
- **Multi-agent architecture.** Evolved a tightly coupled single-agent workflow into three collaborating agents for medical expertise, nursing follow-up and data analysis. The data agent accessed patient records through **MCP/APIs** and calculated time in range (TIR), means and variances; the medical expert agent retrieved authoritative guidelines and produced plain-language explanations; the nursing agent delivered proactive reminders using follow-up SOPs and the heartbeat calendar. Separated responsibilities, tools and knowledge access to reduce coupling between retrieval, numerical analysis and task execution, improving maintainability and interaction reliability.

 Technical Skills

- **LLM training and evaluation:** Transformer architecture; familiarity with Qwen and DeepSeek training methodologies; LoRA and full-parameter SFT, DPO, GRPO and on-policy distillation; model evaluation and failure analysis.
- **Training and inference engineering:** Multi-node, multi-GPU training; GPU memory optimisation; training diagnostics and troubleshooting; model deployment and performance tuning; verl and vLLM.
- **Development workflow and tools:** Proficient in collaborative AI-assisted development, with shared coding conventions and coordinated workflows across concurrent tasks; Linux, Git and Weights & Biases (W&B).
- **Agent frameworks and harnesses:** Proficient with LangChain, LangGraph, AutoGen, Pi, DSH, Codex and CC Harness.

 Publications

27 total citations on Google Scholar.

- *Enhanced Word-Unit Broad Learning System With Sememes*. IEEE Access, IF 4.2. **Second author**.
- *Typhoon Track Prediction Based on TimeForce CNN-LSTM Hybrid Model*. Springer. **First author**.